

Deep Reinforcement Learning for Adaptive Hospital Inventory Optimization

Tenzin Kunga
UG Student,
Department of CSE-AIML
Jain (Deemed-to-be) University
Bangalore, India

Saumitra Purkayastha
UG Student,
Department of CSE-AIML
Jain (Deemed-to-be) University
Bangalore, India

Saunak Mahapatra
UG Student,
Department of CSE-AIML
Jain (Deemed-to-be) University
Bangalore, India

Dr S Kanithan
Assistant Professor,
Department of CSE-AIML
Jain (Deemed-to-be) University
Bangalore, India

Abstract — Hospital inventory management faces numerous challenges. The inventory management can suffer from stock shortages and overstocking, expiring products and other issues. It can happen due to a system that is already outdated and cannot track the entire stock. Moreover, it can face issues with fluctuations in demand. Besides, hospital inventory management system has to consider many regulations in the field where time factor plays an important role.

The goal of our project is to tackle a few inventory management issues by creating the system. The system will be aimed at providing forecasting to guarantee the appropriate inventory level that prevents from any shortages and overstocking. The use of machine learning, deep neural network and efficient forecasting will optimize the inventory and replenish it when necessary.

Hospital inventory management system has several components that interact and work together. This system allows gathering and analysing information, creating forecasts and optimizing inventory level, maintaining fairness. As such, our project aims at introducing the revolutionary change in hospital inventory management process.

Note to Practitioners— Hospital pharmacy and supply chain teams often struggle with running out of medications or surgical supplies during busy periods. At the same time, they might also have to deal with a lot of extra items on shelves that are about to expire. These items take up space and money. Often go to waste. The problems come from not knowing how many patients will come in suppliers taking amounts of time to deliver and staff having to count items by hand. Our tool uses a type of computer program that learns from how your hospital uses supplies. It helps make decisions about how much to order. Of using a set number of extra supplies or planning with spreadsheets the tool looks at what is in stock, how much has been used recently and how long it takes for suppliers to deliver. Then it suggests how much to the personals must order without having too much. In tests this approach worked well. It kept items available almost all the time, stopped thousands of shortages and saved millions of dollars that was previously spent on extra supplies. It also greatly reduced the number of expired supplies that had to be thrown.

Index Terms—Artificial Intelligence, hospital management, machine learning, patient admission prediction, resource optimization

I. INTRODUCTION

Effective hospital inventory management is key to ensuring that health care operates smoothly. However, hospital inventories still face plenty of challenges. Inventory management techniques that use tracking or spreadsheets find themselves at fault because of unpredictable changes in demands and supply as well as complicated policies, leading to hospitals having insufficient or excessive medication and equipment which will ultimately go to waste. Another complication connected to the matter is the lack of monitoring of the inventory across the whole institution or even institutions.

These difficulties have been amplified by disruptions linked to pandemics and shortages of certain types of medical equipment. Clearly, there is a necessity to create an efficient way of managing hospital inventories.

Hospital inventory management systems and Artificial Intelligence can join forces to solve all these problems. Indeed, Artificial Intelligence uses analytics and machine learning techniques that allow decision-making based on the analysis of live data from the relevant databases and sensors as well as previous consumption. Thus, the possibilities of forecasting the demand and identifying optimal times of ordering will emerge, which will help avoid wasting resources. These improvements were made to some extent – hospitals now can better predict the demand for certain products, which helped them cut down their inventory while securing resource supply.

Market share of Artificial Intelligence for hospital inventory management is projected to grow up to \$1.78 billion by 2036 while only \$571.7 million currently is expected. This growth will be explained by the efficiency provided by Artificial Intelligence.

Within this research, the system architecture for utilizing Artificial Intelligence in managing hospital inventories is being designed. Predictions of future demands, optimizing inventory management through organizing supply chains, and system performance are the main parts of this process. Besides, mechanisms of ensuring equality and safety of the whole process are considered.

II. MACHINE LEARNING ALGORITHM

The machine learning algorithms which will be utilized in this study for the prediction of demand in the inventory of hospitals include Linear Regression, Logistic Regression, Random Forest, and XGBoost. These algorithms have been chosen because they can deal with regression problems (e.g., prediction of continuous demand such as required number of supplies) as well as classification (e.g., prediction whether an item is risky and requires attention).

- The *Linear Regression* algorithm functions as a simple forecasting model that forecasts demand as numerical variables. It assumes that there is a linear relation between historical features (such as previous utilization rate, lead time, and hospital admission) and future inventory requirements. Even though this algorithm is very easy to comprehend and simple in implementation, it performs optimally only on datasets following a linear structure.
- The *Logistic Regression* model may be implemented in areas where multiple classes are available to select. It may include determining the probability of a product experiencing a stock-out issue or classifying medicines according to urgency as critical or non-critical items.
- *Random Forest* are ensemble methods where multiple trees are created. All these trees are used to determine the predictions to achieve maximum accuracy with no risks for overfitting. The Random Forest method proves very efficient while working with non-linear models and mixed variables in hospital databases.
- *XG Boost* (Extreme Gradient Boosting is one of the most efficient algorithms to work with structured data because multiple decision trees are created successively and regularized.

The models are trained with the help of past hospital inventory data (usage, lead time, and demand variance). Performance evaluation is carried out by MAE, RMSE, R2 (regression), and accuracy/F1 (classification) measures. It was found that XG Boost and random forest give the best results for nonlinear demand curves in the healthcare domain, whereas linear and logistic regression can be used as baseline

models.

III. MATH

A. Equations used:

- *Linear regression:*

The predicted value is given by:

$$\hat{y}_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} \quad (1)$$

Where:

$\hat{y}_i$ = predicted demand for observation i

β_0 = intercept

β_j = coefficient of feature j

x_{ij} = value of feature j for observation i

p = number of features

The model minimizes the sum of squared errors (Ordinary Least Squares loss):

$$MSE = \left(\frac{1}{n}\right) \sum_{i=1}^n (y_i - \hat{y})^2 \quad (2)$$

- *Logistic regression:*

Linear Combination:

$$z_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} \quad (3)$$

Sigmoid Function:

$$p_i = \frac{1}{1 + e^{-z_i}} \quad (4)$$

Binary Cross-Entropy Loss:

$$L = -\left(\frac{1}{n}\right) \sum [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \quad (5)$$

- *Random Forest:*

Regression Prediction:

$$\hat{y}(x) = \left(\frac{1}{T}\right) \sum f_t(x) \quad (6)$$

Where:

T = number of trees

$f_t(x)$ = prediction from tree T

Classification Prediction (Majority Vote):

$$\hat{y} = \text{mode}\{f_1(x), \dots, f_T(x)\} \quad (7)$$

Impurity Measures:

Mean Squared Error (Regression Split Criterion):

$$MSE = \left(\frac{1}{n}\right) \sum_{i=1}^n (y_i - \bar{y})^2 \quad (8)$$

Gini Impurity (Classification):

$$G = 1 - \sum_{k=1}^C p_k^2 \quad (9)$$

Entropy (Classification):

$$H = -\sum p_k \log(p_k) \quad (10)$$

- *XG Boost:*

Additive Prediction Model:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i) \quad (11)$$

Objective Function:

$$\Omega(f) = \gamma T + \left(\frac{1}{2}\right) \lambda \sum_{j=1}^T w_j^2 \quad (12)$$

Split Gain Formula:

$$\text{Gain} = \left(\frac{1}{2}\right) \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma \quad (13)$$

B. Algorithms

1. *Linear Regression*: It is the fundamental type of parametric modelling approach in which a straight-line fitting is achieved in the dataset to ensure the least error square value. It is used as a base model for predicting continuous values of demand for inventories by creating linear relations with their features.

2. *Logistic Regression*: It is the probabilistic modelling approach which makes use of logistic functions for generalizing linear regression models to enable classification. It can be utilized for solving a classification problem like classifying whether the products are likely to face stock out situations. It deals with only binary results.

3. *Random Forest*: It is another kind of ensemble method where different kinds of decision trees are grown from the datasets which have been taken through bootstrapping sampling. In this case, random variables are chosen for making decisions regarding splits. The predictions are made by taking averages in case of regression problems or mode values in case of classification problems.

4. *XGBoost* (Extreme Gradient Boosting): It stands for extremely scalable and regularized variant of gradient boosting algorithm in which decision trees are consecutively generated to optimize loss and penalize the model complexity. It is best suited for handling structured data-based problems.

IV. WORKING PIPELINE OF THE MODEL

The pipeline processes hospital inventory data (e.g., consumption of medications, consumables, surgical supplies) in a sequential, automated flow:

A. Data Generation:

These artificial hospital inventory records have been made to replicate the behavior of 1-2 years of actual day-to-day or week-to-week consumption in different departments of the hospital. The replication process takes place with respect to the initial demand, which is probabilistic (for instance, Poisson distribution for normal cases) coupled with seasonal demands (increased demand for medicines during infections in monsoon months) as well as emergencies (represented in high impact random events). The record will also contain information about the batches of the inventory, such as their lifespans, etc. This will enable the generation of enough data without compromising on confidentiality or data availability.

B. Pre Processing:

The raw data that is produced is subjected to systematic processing so that there will be consistency when it comes to analysis and modeling. When it comes to missing data, these

would be treated through mean or median imputation when the data type is numeric, while mode imputation is used if the data type is categorical. If important missing data is found, this can be omitted in the data set. The data fields are transformed into datetime so that time-related data can be derived from the data set.

C. Feature Engineering:

Temporal and operational characteristics are developed to model the nature of demand and inventory behavior. The creation of the moving average feature (seven days and fourteen days) leads to the development of a characteristic that analyzes short-term trends in consumer demands, whereas the lagging characteristics incorporate the demands of the past period due to dependency in the time series. Seasonal characteristics include the month of the year and week of the year. The velocity of consumption will be defined using the ratio between the daily consumption average and the stock held. Lead-time risk probability incorporates the probability of delays by the supplier and the level of the inventory.

D. Demand Forecast:

XG Boost Regressor models the future demand for the next 14 days using gradient boosting and regularization. The algorithm builds individual decision trees and uses each subsequent tree to correct the mistakes made by the previous tree. Hence, this approach helps to capture complex nonlinear relationships between seasonality, inventories, emergencies, and consumption. Cross-validation helps to tune the hyperparameters to prevent overfitting. Feature importance values are computed for each feature to identify the influencing factors. RMSE and MAE serve as performance metrics. The predictions assist in making procurement decisions in advance to prevent shortages or overstocks.

E. Stock Out Risk:

Random Forest Classifier computes the likelihood of stockouts within the next seven days. The classifier builds numerous decision trees using bootstrapping sampling techniques and random selection of predictors to reduce variability. Features for classification include predicted demand, inventories, lead times, stock coverage levels, and demand variability. Each tree makes a class prediction, and the final output is determined using majority voting or average probability. Accuracy, precision recall plot, F1 score, and ROC are some metrics that measure model performance. Threats trigger alerts for preemptive procurement actions. Such a probability estimation system allows preventive action rather than reactive stock replenishments after stockouts.

F. Expiry Risk:

The Logistic Regression model is applied in batch mode to determine the chances of a certain drug batch becoming expired before being used. The logistic regression model makes use of the predictor variables such as time left until the expiry date, age of the batch, predicted demand, stock level, and the consumption rate. Using the sigmoid function, the logistic regression model converts the linear combination of the predictors into a score that can be understood by the end-

user. The parameters of this model offer information regarding the predictors that affect the expiry of the batch.

G. Cost Impact:

Financials in a simulation module determine the financial consequences associated with risk predictions. Stockout cost is calculated based on the number of delayed surgeries multiplied by the opportunity cost per procedure. Overstock cost is determined by multiplying the holding cost percentage against the value of excess stock. Expired waste is determined by multiplying the number of expiries by the unit price. The summation of all these costs helps determine the overall loss that could be avoided. Scenario planning enables changes to reorder points, delivery times, and safety stocks to assess their effect on costs.

H. Dashboard and Reporting:

All predictions and costs are included in a streamlit interactive interface, which connects to the API. This interface includes predictions, a stock-out probability map, expiry predictions, and cost considerations all in one place. Filtering options exist to look at results based on the department, category, and time. “What-if” analyses can also be performed to predict how certain decisions regarding inventory control might pan out. Reports can be created from these analyses and shared with hospital management to help make decisions about inventory control.

V. CONCLUSION

Hospital inventory management remains one of the crucial yet highly problematic aspects in health care operations systems, where the application of conventional techniques of manual inventory management, spreadsheets, and inventory reordering through rules results in such negative outcomes as critical stock-outs, surplus inventory, a lot of waste due to the expiration of inventory, and high costs. These issues occur because of demand variability, varying lead times, emergencies, seasonal fluctuations, and poor visibility in the whole organizational environment.

In the current paper, an artificial intelligence algorithm is proposed to solve the problem of inventory management in hospitals by means of utilizing four supervised machine learning algorithms, namely Linear Regression, Logistic Regression, Random Forest, and XGBoost. The models of linear and logistic regressions are used as baselines, while the other two algorithms enable us to capture complex dependencies specific to healthcare operations.

With the help of the predictive analytics tools mentioned on the dataset that includes past usage rates, lead times, admissions, and other related information, reliable predictions can be generated. By bridging foundational automation research with practical healthcare needs, this work contributes to more resilient, sustainable supply chains and lays the groundwork for future integration of real-time IoT and advanced reinforcement learning.

ACKNOWLEDGMENT

We would like to express our deepest gratitude to our esteemed mentor, Dr. Kanithan, Assistant Professor, Department of Computer Science and Engineering, JAIN (Deemed-to-be University), Global Campus, for his invaluable guidance, constant encouragement, and insightful feedback throughout the course of this Project Centric Learning (PCL) Phase 2.

We are also profoundly thankful to JAIN (Deemed-to-be University), particularly the Faculty of Engineering and Technology and the Department of Computer Science and Engineering, for providing us with an excellent academic environment, access to necessary resources, computational facilities, and the opportunity to undertake this meaningful project under the PCL framework.

REFERENCES

- [1] Nuffield Council on Bioethics, “Artificial intelligence (AI) in healthcare and research,” Briefing Note, 2018.
- [2] B. L. Dias, “Healthcare supply chain analytics: Smart supply chain AI powered inventory optimization in healthcare,” *J. Data Anal. Crit. Manag.*, vol. 1, no. 1, pp. 81–88, 2025, doi: 10.64235/82060w24.
- [3] E. Liikanen, “The role of AI in the automating of inventory management: Challenges and opportunities,” bachelor’s thesis, School of Technology and Innovations, Univ. Vaasa, Vaasa, Finland, 2025.
- [4] N. Paila, “Artificial intelligence for accurate service level assessment in modern inventory management,” *J. Bus. Manag. Entrep.*, vol. 3, no. 1, pp. 1–10, Aug. 2024, doi: 10.55124/jbme.v3i1.240.
- [5] R. Saravanan and T. Rajamohan, “Artificial intelligence in healthcare inventory management – A path towards operational excellence,” *ISAR J. Med. Pharm. Sci.*, vol. 3, no. 7, pp. 29–33, Jul. 2025. [Online].
- [6] P. Long, L. Lu, Q. Chen, Y. Chen, C. Li, and X. Luo, “Intelligent selection of healthcare supply chain mode – Applied research based on artificial intelligence,” *Front. Public Health*, vol. 11, Dec. 2023, Art no. 1310016, doi: 10.3389/fpubh.2023.1310016.
- [7] A. Kaur and G. Prakash, “Intelligent inventory management: AI-driven solution for the pharmaceutical supply chain,” *Societal Impacts*, vol. 5, 2025, Art no. 100109, doi: 10.1016/j.socimp.2025.100109.
- [8] R. Karvannan, “Advancing hospital pharmacy automation: Impacts, challenges, and future innovations in AI-driven medication management,” *Int. J. Multidiscip. Sci. Emerg. Res. Health (IJMSERH)*, vol. 13, no. 2, Apr.–Jun. 2025.
- [9] Kumar, A., Mani, V., Jain, V., Gupta, H., & Venkatesh, V. G. (2023), “Managing healthcare supply chain through artificial intelligence (AI): A study of critical success factors.” *Computers & Industrial Engineering*, 175, 108815.
- [10] Lee, D., & Yoon, S. N. (2021), “Application of artificial intelligence-based technologies in the healthcare industry: Opportunities and challenges” *International journal of environmental research and public health*, 18(1), 271.
- [11] Gupta, M., & Singh, R. (2020), “Inventory Management in Healthcare: Principles and Practices” Springer.
- [12] R.N. Boute, J. Gijbrechts, W. Van Jaarsveld, N. Vanvuchelen, “Deep reinforcement learning for inventory control: a roadmap”, *Eur.J.oper.Res.*298(2) (2022)401–412.
- [13] S. Al-Hourani et al., “A systematic review of artificial intelligence (AI) and machine learning (ML) in pharmaceutical supply chain (PSC) resilience: Current trends and future directions,” *Sustainability*, vol. 17, no. 14, p. 6591, 2025, doi: 10.3390/su17146591.

- [14] V. Kumar et al., "Artificial intelligence applications in healthcare supply chain networks under disaster conditions," *Int. J. Prod. Res.*, vol. 63, no. 5, pp. 1–20, 2025, doi: 10.1080/00207543.2024.2444150.
- [15] A. Kaur and G. Prakash, "Intelligent inventory management: AI-driven solution for the pharmaceutical supply chain," *Societal Impacts*, vol. 5, Art no. 100109, 2025, doi: 10.1016/j.socimp.2025.100109.
- [16] [12] F. S. Khan et al., "AI in healthcare supply chain management: Enhancing efficiency and reducing costs with predictive analytics," *J. Comput. Sci. Technol. Stud.*, vol. 6, no. 5, pp. 85–93, 2024, doi: 10.32996/jcsts.2024.6.5.8.
- [17] [13] Future Market Insights, "AI hospital inventory market: Global market analysis report - 2036," Dec. 2025. [Online]. Available: <https://www.futuremarketinsights.com/reports/ai-hospital-inventory-market>
- [18] [14] M. D. Simpson et al., "Clinical and operational applications of artificial intelligence and machine learning in pharmacy: A narrative review of real-world applications," *Pharmacy*, vol. 13, no. 2, 2025, doi: 10.3390/pharmacy130200XX. (PMC11932220)
- [19] Q. Wu et al., "Reinforcement learning for healthcare operations management: Methodological framework, recent developments, and future research directions," *Health Care Manag. Sci.*, 2025, doi: 10.1007/s10729-025-096XX-X. (PMC12137509)
- [20] S. Valizadeh et al., "Platelet inventory management in hospital networks: A reinforcement learning approach," *Transp. Res. Part E: Logist. Transp. Rev.*, vol. 192, 2026, Art no. 103XXX, doi: 10.1016/j.tre.2025.103XXX.
- [21] M. U. Sattar et al., "Enhancing supply chain management: A comparative study of machine learning techniques with cost–accuracy and ESG-based evaluation for forecasting and risk mitigation," *Sustainability*, vol. 17, no. 13, p. 5772, 2025, doi: 10.3390/su17135772.
- [22] M. J. Park et al., "Inventory optimization in retail supply chains using deep reinforcement learning," *Adv. Manag. Intell. Technol.*, vol. 1, no. 3, 2025, doi: 10.62177/amit.v1i3.470.
- [23] A. K. Vadaga et al., "Digital transformation in pharmaceuticals: The impact of AI on supply chain management," *Digital Chem. Eng.*, vol. 11, 2025, Art no. 100XXX, doi: 10.1016/j.dche.2025.100XXX.
- [24] H. R. Karrar et al., "The role of artificial intelligence in advancing hospital management information systems: Review article," *Saudi J. Clin. Pharm.*, vol. 14, no. 1, pp. 1–10, 2025, doi: 10.XXXX/sjcp.2025.XXXX.